Export and Productivity: Distributional Equality Analysis

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Abstract

In this paper, we investigate the theoretical proposition of Melitz regarding the relationships between international trade and labor productivity. We use firm-level data from the UK to compare labor productivity distributions of exporters and non-exporters. Our findings demonstrate that there is a statistically significant difference in labor productivity distributions between exporters and non-exporters across all quantiles.

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1 Introduction

1.1 Overview

International trade relations has been proven to be mutually beneficial over the centuries of economic history. Macroeconomic models have constantly been constructed to reward trade, and countless studies have been published to validate those models empirically. A variable of interest that constantly appeared in simple to complex models was **labor productivity**. Productivity has its roots in the concept of comparative advantage introduced by David Ricardo in 1817. Ricardo's theory of comparative advantage suggested that countries have varying levels of productivity in producing different goods. Countries which have lower opportunity cost for a certain type of good – often measured by hours of labor required to produce one unit of good – will specialize in producing that good and trade with other countries. Subsequent models have built on different aspects of productivity but one of them was groundbreaking in terms of its implications. (Melitz, 2003) Melitz focused on micro data of firms and introduced the concept of firm-level productivity. According to Melitz, firms have different productivity levels within the same industry. Some firms are more productive than others and this property brings with it some important consequences.

Given all agents are rational, the model suggests that comparatively productive firms will be able to cover the fixed costs of entering the export market through efficient generation of revenue. Conversely, firms with lesser productivity would not be able to cover those costs and would be forced to exit the international market. As a result, the average productivity of industries increase as only the more productive firms can afford to operate at an international scale. With a focus on the UK market, our study aims to explore the interactions between international trade and labor productivity in a granular manner and test the theoretical expectations brought by Melitz.

1.2 Hypotheses

Effects of international trade on labor productivity may differ for different subgroups of firms. Although an overall positive relationship has long been agreed upon, the extent/existence of this relationship can be explored further by creating sub-categories of firms. As such, we formed multiple hypotheses to understand empirical presence of the Melitz framework in the UK market.

We test the following hypothesis in our study:

- **H1:** Labor productivity of exporters show statistically significant differences from that of non-exporters in all levels of productivity.
- **H2:** Multinational firms' labor productivity distribution is not same as domestic firms for both exporters and non-exporters.
- **H3:** As firms grow in size, the gap in labor productivity between exporters and non-exporters tends to decrease as larger firms can utilize economies of scale regardless of their market scope. By extension, the labor productivity distribution of exporters and non-exporters approach the same limit for sufficiently large firms.

H1 aims to validate the core premise of the Melitz framework while H2 and H3 are extensions that explore whether Melitz's ideas are unanimously applicable across different firm characteristics.

1.3 Literature Review

There are numerous studies that focus on the relationship between productivity and export status of firms. Empirical work carried out by Bernard and Jensen is one of the remarkable studies in the field investigating the relation between export status and productivity. They use plant-level data from U.S. based manufacturers between 1976 and 1987 to compare the productivity of exporting and non-exporting plants. The study concludes exporting plants to be more productive than non-exporting plants (Bernard and Bradford Jensen, 1999), which aligns with the self-selection hypothesis proposed by Melitz. They also uncover a one-sided causal reationship where productive firms are likely to become expoerters but exporting itself does not necessarily increase productivity. This outcome is aligned with the expectations of Melitz model as well.

Another interesting study has been published by Atkin, Khandelwal, and Osman where a survey among rug producers in Egypt is conducted. Firms participating the experiment are divided into two groups:

- Control Group: Firms that do not receive any assistance to export their products and operate at a domestic scale.
- Treatment Group: Firms that are assisted to export their products internationally through an intermediary.

The experiments starts with both groups being unaware of the export opportunities that will be presented to the treatment group. An intermediary rug firm in Egypt

is used to find buyers in the international market and only the first orders are initiated by the intermediary while the rest is dependent on the satisfaction of the customer and the quality of the rugs produced. Over time, they record an improvement in rug quality produced by the treatment group. By extension, they capture the "learning by doing"[1] effect of exporting through a measurable difference in quality between the treatment pre/post export opportunity. At the end of their experiment they find that the profits of the treatment group increases by 16% to 26% compared to the control group along with the quality improvements of the rugs produced by the treatment group. What makes this study more robust is that at the end of the experiment they gather the control and treatment group and provide them the necessary materials to weave a rug. Treatment group produces much higer quality rugs compared to the control group at the same production time. This shows that exporting has positive effect on treatment groups. Further validation is valuable as profit measures could be biased due to certain affects like product specialization.

2 Data

This study utilizes a dataset of 1,800 entries published by the "Office for National Statistics" (ONS) of the United Kingdom. The dataset focuses on labor productivity through **"Approximate Gross Value Added"** (aGVA) per worker. A granular set of subcategories are provided alongside the ability to easily filter exporters and non-exporters.

Unfortunately, the dataset is not recorded at the firm level. However, the granularity of segmentation create easily accessible and appropriately generalized productivity measures that can be compared directly. Entries follow an annual frequency, spanning from 2005 to 2022.[2]

The data measures quantiles $\hat{Q} = \{Q_{10}, Q_{25}, Q_{50}, Q_{75}, Q_{90}\}$ alongside the mean $\left(\overline{aGVA}\right)$ and the standard deviation (S_{aGVA}). Said measurements are provided for exporters and non-exporters separately. Observations are further segmented by:

- **Size Bands (Employees)**: Small (10-49), Medium (50-249), Large (250+)

[1]The terms refers to the fact that producers gain experience and optimize their processes over time.

[2]The data for 2005-2011 comes from a single source (ABS-TiG) while every data point from 2011 onwards have two separate sources (ABS-TiG and ABS) with entries for each source kept separate.

- **Age Bands (Years)**: New (0-2), Young (3-5), Mature (6-10), Old (10+)
- **Ownership Type**: Domestic, Foreign
- **Industry**: NACE Rev. 2 Categories
- **Region**: 11 UK regions specified by ONS

3 Methodology

All hypotheses of this study concern the possibility of difference in distributions of labor productivity among exporters and non-exporters. Having access to quantile level granularity on a time series basis creates the opportunity to assess the exporter/non-exporter differences in each productivity bracket (Q_i) separately. To transform this data into insightful outcomes, we outlined 3 major steps to construct our empirical analysis:

1. Isolating Empirical Distributions
2. Building the Equality Tests
3. Analysis and Interpretation

3.1 Empirical Distributions

By filtering the dataset, we can create the empirical distributions of productivity brackets to show how each quantile of productivity evolves over time. For any subset of segmentations (e.g., Size Band = Small, Age Band = New), we observe two samples $\{X_{EXP}, X_{NEXP}\}$ for each productivity quantile Q_i. Both sources combined, each categorical group produces $N_{EXP} = N_{NEXP} = 30$ observations.[3]

Using this filtering method the following distributions will be built:

- Quantile distributions of Exporters/Non-Exporters without categorical filtering.
- Quantile distributions of Foreign and Domestic owned Exporters/Non-Exporters.
- Distribution of Median (Q_{50}) Productivity for Exporters/Non-Exporters by Size Band.

NOTE: "Quantile distribution" refers to the comparison of each quantile Q_i separately over time between Exporters and Non-Exporters. (i.e. $Q_{[i,\ 2005:2022]} \forall Q_i \in \hat{Q}$ for Exporters vs. Non-Exporters)

[3] 30 observations assuming all data for all years are available. This is not always the case. Refer to Appendix 8.1 and 8.2 for a detailed outlook on the missing data points.

3.2 Equality Tests

Having built the empirical distributions, we can compare their equality using the two-sample **Kolmogorov-Smirnov (KS)** test.[4] The KS test utilizes a simple statistic $D = \sup_x |F_i(x) - F_j(x)|$ to measure the maximum distance between two empirical CDFs. By the outcomes of each KS test, we can conclude that the tested samples "share" the same distribution.

3.3 Analysis and Interpretation

Applying the KS test to all constructed empirical distributions will yield a series of p-values. By analyzing these p-values, we can accept or reject the null hypothesis of equality of distributions for each productivity quantile and categorical segmentation; potentially allowing us to discover empirical quirks that contradict the Melitz framework. Of course, it is important to note that any such discoveries will be confined to the surveyed portion of the UK economy and our paper does not aim to challenge the Meliz expectations at a theoretical level if such findings occur.

With the specific set of hypotheses being tested, we expect to see sub-segmentations where exporters' and non-exporters' sample distributions are statistically indifferent. Therefore, a stricter, two-sided equality test is preferred. We accept the assumption that Exporters have greater productivity where a significant difference of distributions is observed.[5]

4 Results

The first course of action is the verification of H1. Therefore, we compare the aGVA observations of all quantile levels for exporters and non-exporters.

[4]Refer to Section 7.1 for further discussion regarding the KS test assumptions and how they interact with our data.

[5]We will numerically display that Exporters have greater productivity in these cases, but we will not provide a robust and statistically supported proof.

Q_i	p-value	Equality Conclusion
Q_{10}	$\approxeq 0$	Reject H_0
Q_{25}	$\approxeq 0$	Reject H_0
Q_{50}	$\approxeq 0$	Reject H_0
Q_{75}	$\approxeq 0$	Reject H_0
Q_{90}	$\approxeq 0$	Reject H_0

Table 1: KS Test Results: Exporters vs. Non-Exporters (All Firms)

As expected, and demonstrated in literature, we observe a strong statistical evidence of productivity difference in the UK market. However, seeing a significant difference with all categories included does not guarantee a difference will be observed when sub-categories are tested separately.

We then explore the productivity differences between domestic and foreign-owned (not necessarily multinational) firms' export status. For all quantiles, foreign/domestic owned exporters and non-exporters are compared for equality.

p-values for Exporters vs. Non-Exporters by Ownership Type

	Foreign-Owned	Domestic-Owned
Q_{10}	$\approxeq 0$	$\approxeq 0$
Q_{25}	$\approxeq 0$	$\approxeq 0$
Q_{50}	$\approxeq 0$	$\approxeq 0$
Q_{75}	0.015643	$\approxeq 0$
Q_{90}	0.594071	$\approxeq 0$

Table 2: KS Test Results: Exporters vs. Non-Exporters by Ownership Type

Although most tests result in a strong rejection, we see an interesting result in the foreign-owned firms. As we cross Q_{75}, the p-values start increasing ultimately leading to a Q_{90} where we fail to reject the equality of distributions. In the UK data, we can therefore conclude that foreign-owned companies at the upper echelon of labor productivity do not exhitibit an increased productivity tied to their export status.

Finally, we test H3 by comparing exporters/non-exporters of each size band at all quantile levels. Under H3, we expect to see larger p-values as we move towards larger size bands and higher quantile brackets.

p-values for Exporters vs. Non-Exporters by Size Band

	Micro (0-9)	Small (10-49)	Medium (50-249)	Large (250+)
Q_{10}	0.001116*	$\approxeq 0$	$\approxeq 0$	$\approxeq 0^*$
Q_{25}	$\approxeq 0^*$	$\approxeq 0$	$\approxeq 0$	$\approxeq 0^*$
Q_{50}	$\approxeq 0^*$	$\approxeq 0$	$\approxeq 0$	$\approxeq 0^*$
Q_{75}	$\approxeq 0^*$	$\approxeq 0$	$\approxeq 0$	$\approxeq 0^*$
Q_{90}	$\approxeq 0^*$	$\approxeq 0$	$\approxeq 0$	$\approxeq 0^*$

* Both samples being compared have a small sample size ($n_i = n_j = 20$). The p-value is included but should not be treated as accurate. Refer to the raw observations in Appendix 8.2. Visual observation aligns with the test results but the p-values themselves are unconfirmable.

Table 3: KS Test Results: Exporters vs. Non-Exporters by Size Band

Contrary to our expectations, we saw a strict and strong rejection in all sizes and quantile levels. This shows that the positive relationship between export status and labor productivity is persistent across all sizes of firms in the UK market.

Seeing the result of the H3 tests, we further explored the relationship between size and productivity within each export status. We compared the median (Q_{50}) of size bands within exporters and non-exporters separately to see if growth in size leads to statistically significant increases in productivity.

p-values for Size Band Comparisons within Exporters

	Micro (0-9)	Small (10-49)	Medium (50-249)	Large (250+)
Micro (0-9)	1	$\approxeq 0$	$\approxeq 0$	$\approxeq 0^*$
Small (10-49)	$\approxeq 0$	1	0.392945	0.001347
Medium (50-249)	$\approxeq 0$	0.392945	1	0.034520
Large (250+)	$\approxeq 0^*$	0.001347	0.034520	1

p-values for Size Band Comparisons within Non-Exporters

	Micro (0-9)	Small (10-49)	Medium (50-249)	Large (250+)
Micro (0-9)	1	0.399477	0.000008	0.983137*
Small (10-49)	0.399477	1	0.000001	0.678914
Medium (50-249)	0.000008	0.000001	1	0.000055
Large (250+)	0.983137*	0.678914	0.000055	1

* Refer to the footnote on Table 3 regarding p-values marked with "$*$".

Table 4: KS Test Results: Size Band Comparisons within Exporters and Non-Exporters.

We observe that most size combinations lead to a rejection. However, both exporters and non-exporters display some size bands that do not exhibit statistically significant productivity differences. Specifically we see that the Small-Medium pair in exporters show similar productivity distributions while the Micro-Large, Micro-Small, and Small-Large pairs in non-exporters do not exhibit significant differences.

5 Conclusion

Through-out this study, our aim was to investigate the results of the framework introduced by Melitz. We believe that our findings provide beneficial insights into the casual relationship between firm productivity and international trade. Our results verifies the self selection hypothesis of Meliz by showing that labor productivity distribution of exporters is significantly different from that of non-exporters across all levels of quantiles. Moreover, we find that within exporters and nonexporter, firm-size plays a cruical role in determining the labor productivity distribution of firms. Albeit, we don't put forward a conclsuive evidence such as elasticity estimates. Further research can be conducted to estimate the magnitue of the effects in more depth by employing different methodologies. Additionally, our study is limited to the UK market due to data availability. Future studies can be carried out on firm level data from different countries to see whether the findings hold true in different economic contexts. Overall, we find it sufficient to conclude the differnece of productivty setting between exporters and nonexports in simple yet effective manner.

6 Bibliography

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7 Remarks

7.1 On the Application of the KS Test

Our data has two properties that creates potential violations of the KS assumptions:

- **Dependence within Samples**: Having time-dependent observations within each sample, it is fair to expect some level of autocorrelation in-sample. If such autocorrelation exists, our samples would no longer be **i.i.d.** thus rendering the KS test invalid.

- **Dependence between Samples**: Our samples are drawn from the same time periods, and slow-moving trends act on productivity in the same ways for exporters and non-exporters. For example, technological advancements (A) increases labor productivity for both exporters and non-exporters over time. In a shared inter-sample macroeconomic environment, there will likely be some level of correlation between the samples as well. As well as putting the i.i.d. assumption on shaky grounds, this has the potential to create inter-sample interactions that the KS test is not designed to account for.

To assure brevity and remaining within the page count limitations, we elected to assume that the KS test is applicable to all samples evaluated. We understand that this would very likely be undefensible as an empirical analysis methodology. With no such limitations, could have elected to define a Block Bootstrap KS test or move further with alternative non-parametric tests.

8 Appendix

8.1 Firm Counts per Subcategories

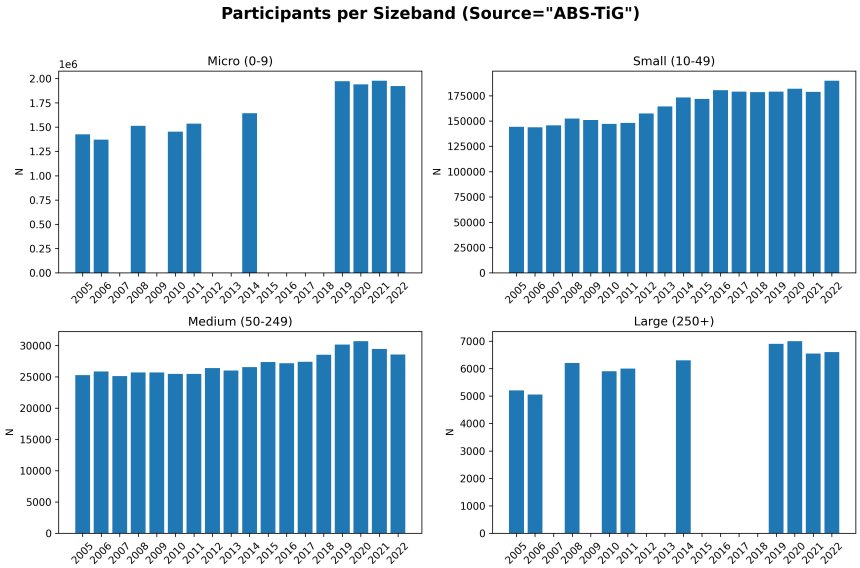

Figure 1: Participants per Sizeband (Source="ABS-TiG")

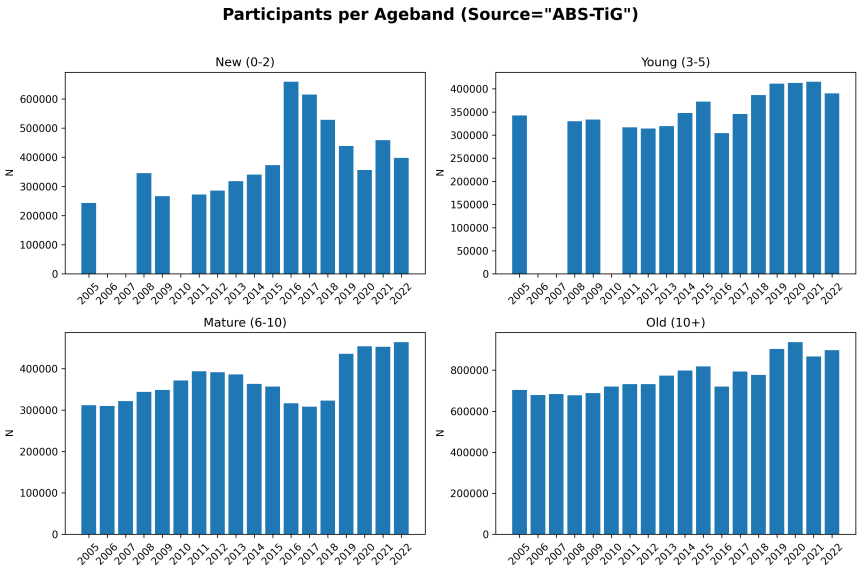

Figure 2: Participants per Ageband (Source="ABS-TiG")

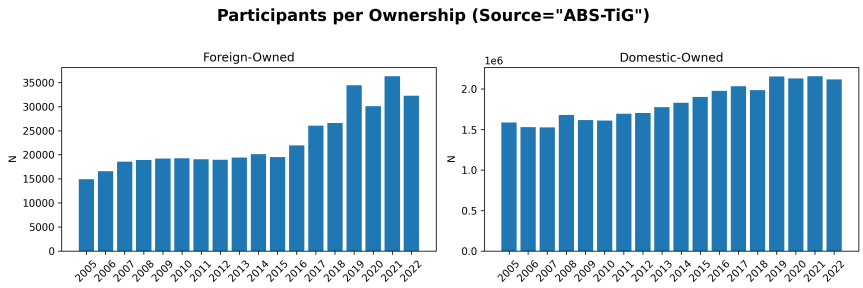

Figure 3: Participants per Ownership Type (Source="ABS-TiG")

8.2 Raw Data Empirical Distributions

All Firms

Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	9000	26000	54000	98000	180500
2006 (ABS-TiG)	9000	27500	54500	100500	175500
2007 (ABS-TiG)	6000	30500	60500	109000	178000
2008 (ABS-TiG)	7500	27500	53500	99000	173500
2009 (ABS-TiG)	6500	24500	50500	87500	154000
2010 (ABS-TiG)	5500	24000	53000	94500	165000
2011 (ABS-TiG)	8500	29000	56500	99500	165500
2011 (ABS)	4500	23000	51000	92000	153000
2012 (ABS-TiG)	8000	27000	54500	97500	169500
2012 (ABS)	4000	22500	48500	90000	155500
2013 (ABS-TiG)	9500	27000	54500	98000	172500
2013 (ABS)	5000	21500	49000	92500	162500
2014 (ABS-TiG)	11000	30500	58000	104500	183000
2014 (ABS)	7000	26500	54500	99500	171500
2015 (ABS-TiG)	9500	28000	57000	98500	166000
2015 (ABS)	6500	23500	52000	93000	154000
2016 (ABS-TiG)	6500	22000	50000	91500	146000
2016 (ABS)	6000	21500	49000	94000	155500
2017 (ABS-TiG)	13000	34500	54500	95000	162000
2017 (ABS)	8500	27500	49500	89000	152000
2018 (ABS-TiG)	9000	26500	48500	96000	155000
2018 (ABS)	3500	22000	43500	88500	146000
2019 (ABS-TiG)	9000	28000	52000	91000	157500
2019 (ABS)	7000	24000	47500	85000	153000
2020 (ABS-TiG)	7000	23000	48500	79500	128000
2020 (ABS)	6000	18000	41000	72000	126500
2021 (ABS-TiG)	3000	21500	44500	89000	159500
2021 (ABS)	2000	18000	40500	80000	151000
2022 (ABS-TiG)	11000	29000	56500	99500	160000
2022 (ABS)	8000	24000	50500	91500	156000

Non-Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	4000	15500	34500	64500	112500
2006 (ABS-TiG)	3500	16500	36000	68500	117000
2007 (ABS-TiG)	3500	15500	36000	68500	118000
2008 (ABS-TiG)	2000	13500	33500	63500	110500
2009 (ABS-TiG)	2500	12500	30000	57500	103000
2010 (ABS-TiG)	2500	13000	31000	60000	107500
2011 (ABS-TiG)	2500	13500	31500	60500	107500
2011 (ABS)	2500	14000	31500	61500	108000
2012 (ABS-TiG)	3000	14000	32000	62000	109000
2012 (ABS)	3500	14000	32000	63000	110000
2013 (ABS-TiG)	3500	14000	33500	64500	110500
2013 (ABS)	4000	14500	34000	64500	111000
2014 (ABS-TiG)	4000	14500	33500	66000	116000
2014 (ABS)	4500	14500	34000	66500	117000
2015 (ABS-TiG)	4500	15000	33500	65500	113000
2015 (ABS)	4500	15000	34000	66000	114000
2016 (ABS-TiG)	3500	13000	29500	58000	98000
2016 (ABS)	4000	13500	29500	58000	97000
2017 (ABS-TiG)	5000	14000	31500	61500	100500
2017 (ABS)	5000	14000	32000	61500	100000
2018 (ABS-TiG)	4500	14000	30000	58000	103000
2018 (ABS)	5000	14000	30500	58000	103000
2019 (ABS-TiG)	3000	12000	28000	56000	100500
2019 (ABS)	3000	12000	28500	56500	102500
2020 (ABS-TiG)	1500	9500	24000	48500	89000
2020 (ABS)	1500	9500	25000	49000	87500
2021 (ABS-TiG)	1000	10000	26000	51000	93000
2021 (ABS)	1500	10000	26000	53000	95000
2022 (ABS-TiG)	3000	12000	29000	57000	96000
2022 (ABS)	3000	12500	29500	58000	98000

Figure 4: Empirical Distributions of Exporters vs. Non-Exporters over Years and Sources

Micro Firms (0-9 Employees)

Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	6000	23000	54000	104000	194500
2006 (ABS-TiG)	5500	23000	52500	101000	185000
2007 (ABS-TiG)	-	-	-	-	-
2008 (ABS-TiG)	4500	24000	52000	99000	177000
2009 (ABS-TiG)	-	-	-	-	-
2010 (ABS-TiG)	4000	19500	48500	94000	168500
2011 (ABS-TiG)	6500	24500	53500	99000	166500
2011 (ABS)	2500	18000	45500	89500	152500
2012 (ABS-TiG)	-	-	-	-	-
2012 (ABS)	2500	18000	45000	88500	158500
2013 (ABS-TiG)	-	-	-	-	-
2013 (ABS)	3000	17500	45000	90000	164500
2014 (ABS-TiG)	8500	26500	55500	104000	185500
2014 (ABS)	5000	23000	51500	97000	174000
2015 (ABS-TiG)	-	-	-	-	-
2015 (ABS)	5000	20000	48000	90500	154000
2016 (ABS-TiG)	-	-	-	-	-
2016 (ABS)	3000	17000	44000	91500	155000
2017 (ABS-TiG)	-	-	-	-	-
2017 (ABS)	-	-	-	-	-
2018 (ABS-TiG)	-	-	-	-	-
2018 (ABS)	-	-	-	-	-
2019 (ABS-TiG)	7000	23500	51000	89000	165500
2019 (ABS)	4500	23000	45000	82500	154500
2020 (ABS-TiG)	5500	21000	47000	74000	119500
2020 (ABS)	4500	17000	35500	68500	119500
2021 (ABS-TiG)	2000	17500	40000	80000	158500
2021 (ABS)	1000	16000	36000	70000	147000
2022 (ABS-TiG)	10000	27000	52500	99000	158500
2022 (ABS)	8000	20000	46000	88000	150000

Non-Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	3500	15500	34500	65000	114500
2006 (ABS-TiG)	3000	16000	36000	69500	119000
2007 (ABS-TiG)	-	-	-	-	-
2008 (ABS-TiG)	1500	13500	33500	64000	113000
2009 (ABS-TiG)	-	-	-	-	-
2010 (ABS-TiG)	2000	12500	30500	60500	110000
2011 (ABS-TiG)	2000	13000	31000	61000	108000
2011 (ABS)	2500	13500	31500	61500	108500
2012 (ABS-TiG)	-	-	-	-	-
2012 (ABS)	3000	14000	32000	64000	111500
2013 (ABS-TiG)	-	-	-	-	-
2013 (ABS)	3500	14500	34000	65500	113000
2014 (ABS-TiG)	4000	14500	33500	67000	118000
2014 (ABS)	4000	14500	34000	68000	119000
2015 (ABS-TiG)	-	-	-	-	-
2015 (ABS)	4500	15000	34000	67000	115000
2016 (ABS-TiG)	-	-	-	-	-
2016 (ABS)	3500	13000	29500	57500	97000
2017 (ABS-TiG)	-	-	-	-	-
2017 (ABS)	-	-	-	-	-
2018 (ABS-TiG)	-	-	-	-	-
2018 (ABS)	-	-	-	-	-
2019 (ABS-TiG)	2500	11500	27500	56000	101000
2019 (ABS)	2500	11500	28000	57000	104000
2020 (ABS-TiG)	1000	9500	24000	48000	89000
2020 (ABS)	1000	9500	25000	49000	87500
2021 (ABS-TiG)	1000	10000	25500	50500	92000
2021 (ABS)	1000	10000	26000	51500	94000
2022 (ABS-TiG)	2500	12000	28500	57000	96500
2022 (ABS)	2500	12000	29000	58000	98000

Figure 5: Empirical Distributions of Micro Size Exporters vs. Non-Exporters over Years and Sources

Small Firms (10-49 Employees)

Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	18500	31500	54000	88500	152000
2006 (ABS-TiG)	19500	35000	60000	102000	161000
2007 (ABS-TiG)	21500	40000	65000	100000	160000
2008 (ABS-TiG)	18000	36500	58000	98000	166500
2009 (ABS-TiG)	15000	31500	54500	89000	146000
2010 (ABS-TiG)	16500	35500	60000	96500	158000
2011 (ABS-TiG)	23000	40000	64500	104000	164500
2011 (ABS)	21000	37500	61500	97500	156000
2012 (ABS-TiG)	19000	36000	59500	97000	156500
2012 (ABS)	18500	35000	57500	91500	149500
2013 (ABS-TiG)	18500	37500	61500	100500	166000
2013 (ABS)	17500	36000	59500	96000	157500
2014 (ABS-TiG)	21000	39500	64000	104500	172500
2014 (ABS)	20500	38000	62500	102000	165500
2015 (ABS-TiG)	16500	37000	62500	101000	159500
2015 (ABS)	16000	35500	61500	99500	154000
2016 (ABS-TiG)	18500	37500	61500	102500	165000
2016 (ABS)	20500	37000	58500	98000	153500
2017 (ABS-TiG)	21500	38500	62500	102500	161000
2017 (ABS)	21000	38000	59500	100000	156000
2018 (ABS-TiG)	20500	36000	59500	99500	147000
2018 (ABS)	20500	35000	57000	96000	144000
2019 (ABS-TiG)	18500	36500	59000	93000	145500
2019 (ABS)	19000	35000	56500	91500	144500
2020 (ABS-TiG)	15000	32000	54000	89000	147500
2020 (ABS)	14500	32000	54000	83500	138500
2021 (ABS-TiG)	17500	38000	62500	102000	165500
2021 (ABS)	14500	37000	62500	101000	165500
2022 (ABS-TiG)	17500	35500	64000	103500	161500
2022 (ABS)	16500	35000	62500	101500	158500

Non-Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	8500	18500	35000	58000	91000
2006 (ABS-TiG)	8000	19000	38000	60500	96500
2007 (ABS-TiG)	8000	19000	37500	62500	102000
2008 (ABS-TiG)	7000	16500	34000	56000	89500
2009 (ABS-TiG)	6500	15500	31500	52000	83500
2010 (ABS-TiG)	6000	16500	33000	55000	89000
2011 (ABS-TiG)	6500	17000	34000	57000	96000
2011 (ABS)	6500	17500	35000	58500	99500
2012 (ABS-TiG)	7000	16500	33000	55500	87000
2012 (ABS)	7000	16000	32500	55000	88500
2013 (ABS-TiG)	6500	16000	32000	56500	93000
2013 (ABS)	6500	16000	31500	56500	93500
2014 (ABS-TiG)	6500	14500	30000	55500	91500
2014 (ABS)	6500	15000	31000	56000	95000
2015 (ABS-TiG)	6500	15000	31000	57000	91500
2015 (ABS)	6500	15000	31500	57500	93500
2016 (ABS-TiG)	7000	16000	33000	60000	97500
2016 (ABS)	7000	15500	33000	60500	99000
2017 (ABS-TiG)	6500	15000	31500	58000	94000
2017 (ABS)	6500	14500	31000	58000	94500
2018 (ABS-TiG)	7000	14500	29000	54000	86000
2018 (ABS)	7000	14500	29000	54500	86500
2019 (ABS-TiG)	6000	14000	30500	55000	89500
2019 (ABS)	6000	14000	30500	55500	90500
2020 (ABS-TiG)	2000	8500	25000	50000	85500
2020 (ABS)	2500	9000	25500	50000	88000
2021 (ABS-TiG)	3000	11000	28000	57500	97500
2021 (ABS)	3500	11500	28500	58500	99500
2022 (ABS-TiG)	5000	12500	31000	55500	93000
2022 (ABS)	5500	13500	32000	57500	96000

Figure 6: Empirical Distributions of Small Size Exporters vs. Non-Exporters over Years and Sources

Medium Firms (50-249 Employees)

Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	17000	31500	51000	85000	144000
2006 (ABS-TiG)	19500	35500	57500	91000	154000
2007 (ABS-TiG)	21000	37500	60000	99500	166500
2008 (ABS-TiG)	20000	36500	59500	96500	151500
2009 (ABS-TiG)	16000	31000	52500	84000	137000
2010 (ABS-TiG)	21000	36500	57500	90500	150000
2011 (ABS-TiG)	22500	40000	61000	95500	153500
2011 (ABS)	22500	39500	60500	91500	148000
2012 (ABS-TiG)	21000	39500	61000	97500	155500
2012 (ABS)	21500	38000	59500	94500	147000
2013 (ABS-TiG)	23000	39000	63000	100000	159500
2013 (ABS)	22500	39000	62500	98000	155500
2014 (ABS-TiG)	23000	40000	64500	109500	182500
2014 (ABS)	24000	40000	64000	103000	167000
2015 (ABS-TiG)	25000	40500	65000	104500	158500
2015 (ABS)	24500	39500	63000	103000	156000
2016 (ABS-TiG)	23500	40000	62500	100500	159500
2016 (ABS)	24000	40500	62000	99500	156500
2017 (ABS-TiG)	23500	40500	63500	103500	167500
2017 (ABS)	24000	40500	63500	102500	165000
2018 (ABS-TiG)	25000	40500	62000	97500	158000
2018 (ABS)	25000	40000	61500	96000	151500
2019 (ABS-TiG)	26000	41500	62000	99000	152000
2019 (ABS)	26000	41000	61000	94500	144000
2020 (ABS-TiG)	18000	35500	58000	92000	146000
2020 (ABS)	19000	35500	57000	89000	140000
2021 (ABS-TiG)	22000	43500	68000	109000	171000
2021 (ABS)	22500	43500	68000	108000	169000
2022 (ABS-TiG)	20000	42000	67500	107000	167000
2022 (ABS)	21500	42500	68000	106500	164500

Non-Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	8000	18500	36000	58500	91000
2006 (ABS-TiG)	7000	17500	35500	59500	90000
2007 (ABS-TiG)	7500	18500	36000	60500	92500
2008 (ABS-TiG)	6500	18000	35000	56500	87500
2009 (ABS-TiG)	7000	16500	32500	53000	83500
2010 (ABS-TiG)	7000	17000	33500	55500	84500
2011 (ABS-TiG)	6000	16000	33000	56000	90000
2011 (ABS)	5500	16500	35000	58000	95000
2012 (ABS-TiG)	7000	17500	35500	58000	99500
2012 (ABS)	6500	18000	37000	62000	105000
2013 (ABS-TiG)	8500	19000	36500	60500	96500
2013 (ABS)	8500	19500	38000	62000	101500
2014 (ABS-TiG)	8000	18000	36500	63500	100000
2014 (ABS)	8000	19500	39000	66500	111000
2015 (ABS-TiG)	7500	18000	35000	62000	109500
2015 (ABS)	7000	18500	37000	66000	113000
2016 (ABS-TiG)	8500	18500	34500	59500	99000
2016 (ABS)	8500	18500	35500	61000	103000
2017 (ABS-TiG)	9000	18500	35500	59000	95000
2017 (ABS)	9000	18500	36500	60500	97500
2018 (ABS-TiG)	9000	19000	34000	56500	89500
2018 (ABS)	9000	19500	35000	58000	94500
2019 (ABS-TiG)	7500	17000	34000	55000	83500
2019 (ABS)	7500	18000	35000	56500	88500
2020 (ABS-TiG)	2500	11000	27500	51000	83500
2020 (ABS)	3000	11500	29000	53500	89000
2021 (ABS-TiG)	6000	16000	35500	63500	102500
2021 (ABS)	6000	16500	37000	66500	106000
2022 (ABS-TiG)	7000	17500	37000	63500	101500
2022 (ABS)	7000	18000	38500	66500	106000

Figure 7: Empirical Distributions of Medium Size Exporters vs. Non-Exporters over Years and Sources

Large Firms (250+ Employees)

Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	17000	31500	52500	89000	149000
2006 (ABS-TiG)	17500	34500	58500	99000	173500
2007 (ABS-TiG)	-	-	-	-	-
2008 (ABS-TiG)	15000	32500	58000	99500	166000
2009 (ABS-TiG)	-	-	-	-	-
2010 (ABS-TiG)	16000	34500	59500	101500	179000
2011 (ABS-TiG)	12500	32000	58000	101500	173500
2011 (ABS)	18500	37000	64500	107000	176500
2012 (ABS-TiG)	-	-	-	-	-
2012 (ABS)	19000	37000	64500	107500	182000
2013 (ABS-TiG)	-	-	-	-	-
2013 (ABS)	20000	39000	67000	109500	178500
2014 (ABS-TiG)	17000	35500	64500	108000	180500
2014 (ABS)	22500	40500	70000	114000	183500
2015 (ABS-TiG)	-	-	-	-	-
2015 (ABS)	24000	41000	68000	110500	184500
2016 (ABS-TiG)	-	-	-	-	-
2016 (ABS)	24500	41500	67000	107500	175500
2017 (ABS-TiG)	-	-	-	-	-
2017 (ABS)	-	-	-	-	-
2018 (ABS-TiG)	-	-	-	-	-
2018 (ABS)	-	-	-	-	-
2019 (ABS-TiG)	14000	32500	56000	97500	163000
2019 (ABS)	23500	41000	64500	105500	172000
2020 (ABS-TiG)	9000	26000	53000	95500	159000
2020 (ABS)	14500	35000	63500	104000	165500
2021 (ABS-TiG)	15000	35500	65500	113000	188500
2021 (ABS)	22000	43000	71500	120500	191000
2022 (ABS-TiG)	17500	37000	66500	112000	183500
2022 (ABS)	24000	45500	72000	116500	181500

Non-Exporters

	p10	p25	p50	p75	p90
2005 (ABS-TiG)	6000	13500	30500	61000	124500
2006 (ABS-TiG)	6000	14000	29000	55000	89000
2007 (ABS-TiG)	-	-	-	-	-
2008 (ABS-TiG)	6500	13500	27500	53500	92500
2009 (ABS-TiG)	-	-	-	-	-
2010 (ABS-TiG)	7000	15000	28500	54000	101500
2011 (ABS-TiG)	6500	14500	27500	54000	88500
2011 (ABS)	7000	16500	34000	60500	109500
2012 (ABS-TiG)	-	-	-	-	-
2012 (ABS)	7500	15500	32500	60000	107000
2013 (ABS-TiG)	-	-	-	-	-
2013 (ABS)	8000	16500	33500	64000	113000
2014 (ABS-TiG)	7000	15000	28000	55000	97000
2014 (ABS)	7500	17000	34000	65000	119000
2015 (ABS-TiG)	-	-	-	-	-
2015 (ABS)	8000	17500	32000	63500	116000
2016 (ABS-TiG)	-	-	-	-	-
2016 (ABS)	7000	16000	31500	58000	107500
2017 (ABS-TiG)	-	-	-	-	-
2017 (ABS)	-	-	-	-	-
2018 (ABS-TiG)	-	-	-	-	-
2018 (ABS)	-	-	-	-	-
2019 (ABS-TiG)	8500	17500	30500	56000	100000
2019 (ABS)	9000	18500	34500	61500	116500
2020 (ABS-TiG)	4500	12500	24500	48500	86000
2020 (ABS)	5000	13500	28000	54000	105000
2021 (ABS-TiG)	6500	15000	30000	58500	104000
2021 (ABS)	7500	18000	35000	70000	122500
2022 (ABS-TiG)	9000	17500	31500	59500	106000
2022 (ABS)	9500	20000	36500	69000	127000

Figure 8: Empirical Distributions of Large Size Exporters vs. Non-Exporters over Years and Sources